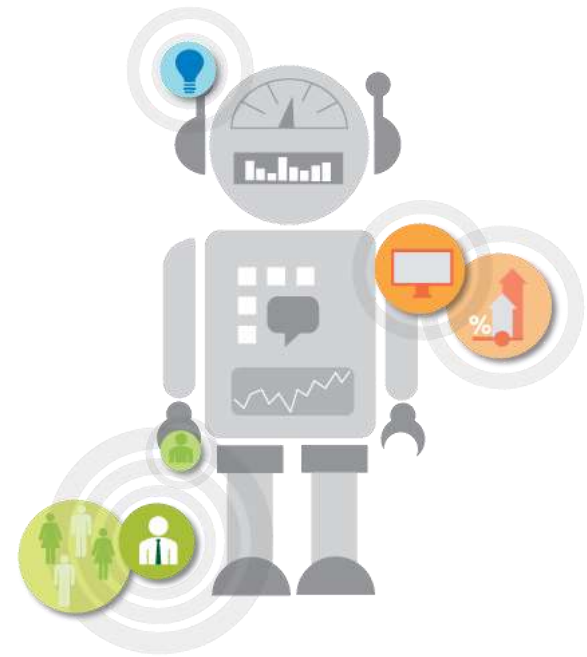


AI and Society

CSET-485

Dr. Pulkit Dwivedi



UNIT - 3



Hierarchical Clustering

Hierarchical Clustering

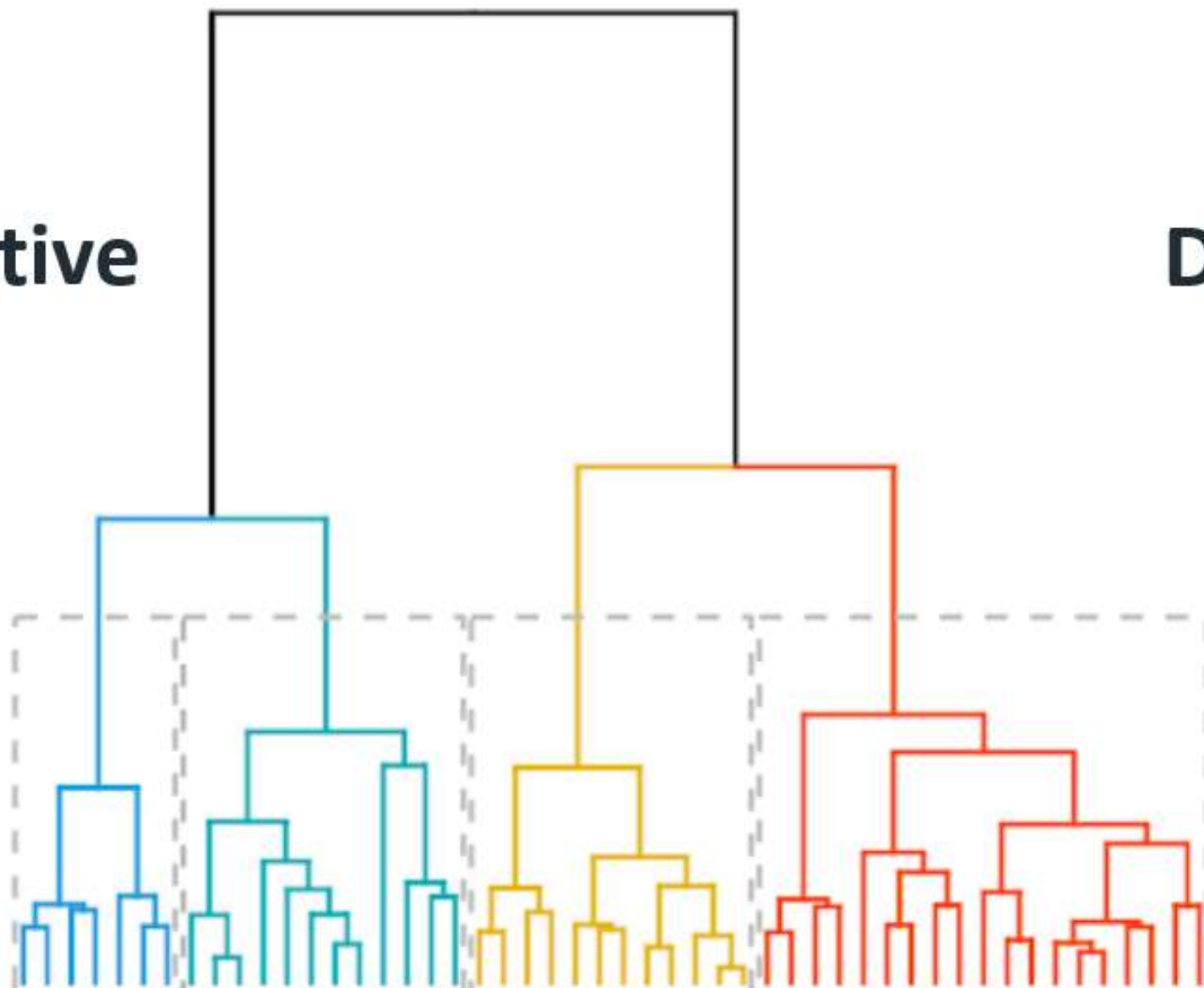
- Hierarchical Clustering creates clusters in a hierarchical tree-like structure (also called a Dendrogram).
- A subset of similar data is created in a tree-like structure in which the root node corresponds to entire data, and branches are created from the root node to form several clusters.
- Hierarchical Clustering is of two types.
 - **Divisive**
 - **Agglomerative Hierarchical Clustering**

Hierarchical Clustering

- Divisive Hierarchical Clustering is also termed as a top-down clustering approach. In this technique, entire data or observation is assigned to a single cluster. The cluster is further split until there is one cluster for each data or observation.
- Agglomerative Hierarchical Clustering is popularly known as a bottom-up approach, wherein each data or observation is treated as its cluster. A pair of clusters are combined until all clusters are merged into one big cluster that contains all the data.
- Both algorithms are exactly the opposite of each other.

Agglomerative

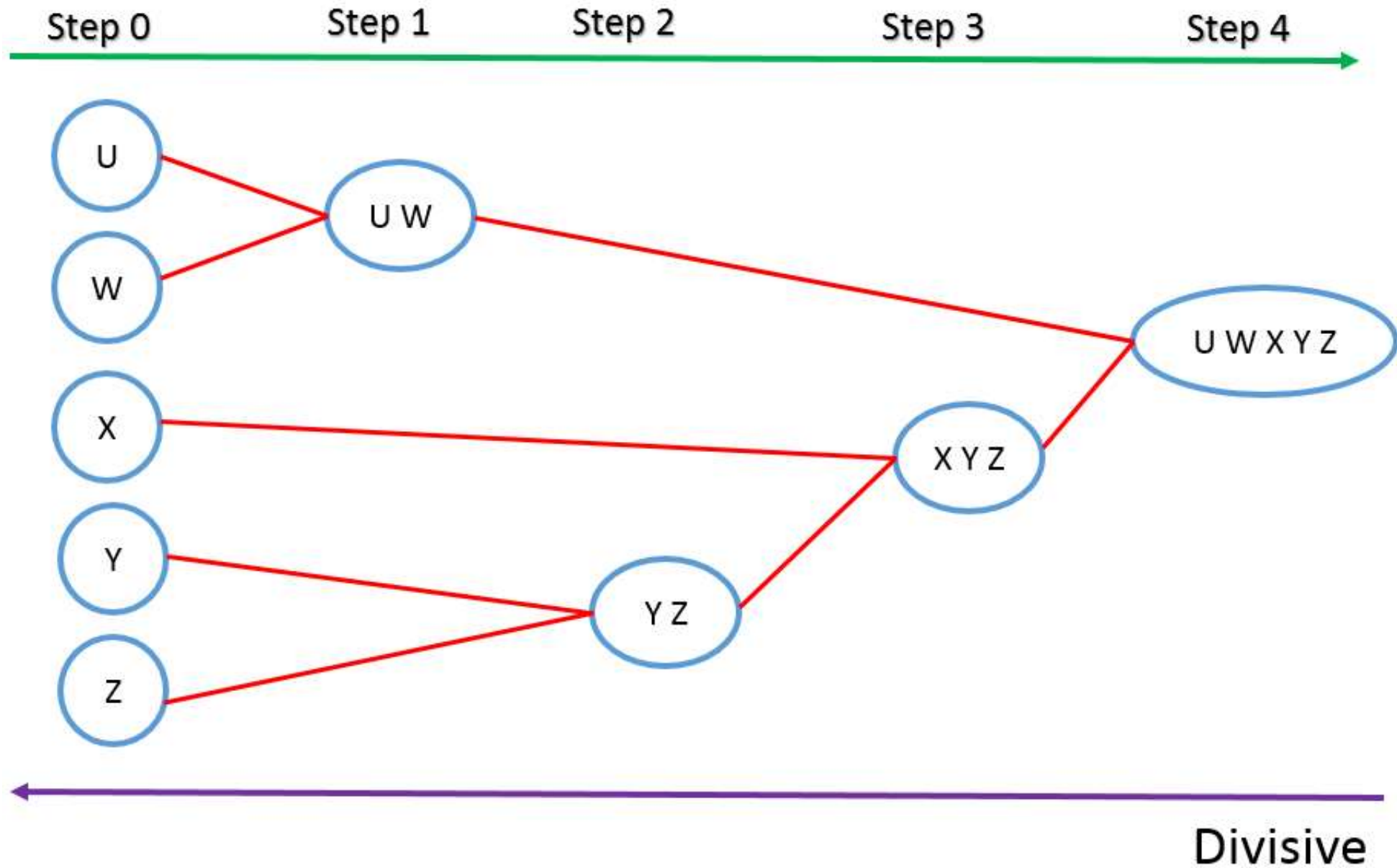
Divisive



Agglomerative Hierarchical Clustering

- For a set of N observations to be clustered:
 1. Start assigning each observation as a single point cluster, so that if we have N observations, we have N clusters, each containing just one observation.
 2. Find the closest (most similar) pair of clusters and make them into one cluster, we now have $N-1$ clusters. This can be done in various ways to identify similar and dissimilar measures.
 3. Find the two closest clusters and make them to one cluster. We now have $N-2$ clusters. This can be done using agglomerative clustering linkage techniques.
 4. Repeat steps 2 and 3 until all observations are clustered into one single cluster of size N .

Agglomerative

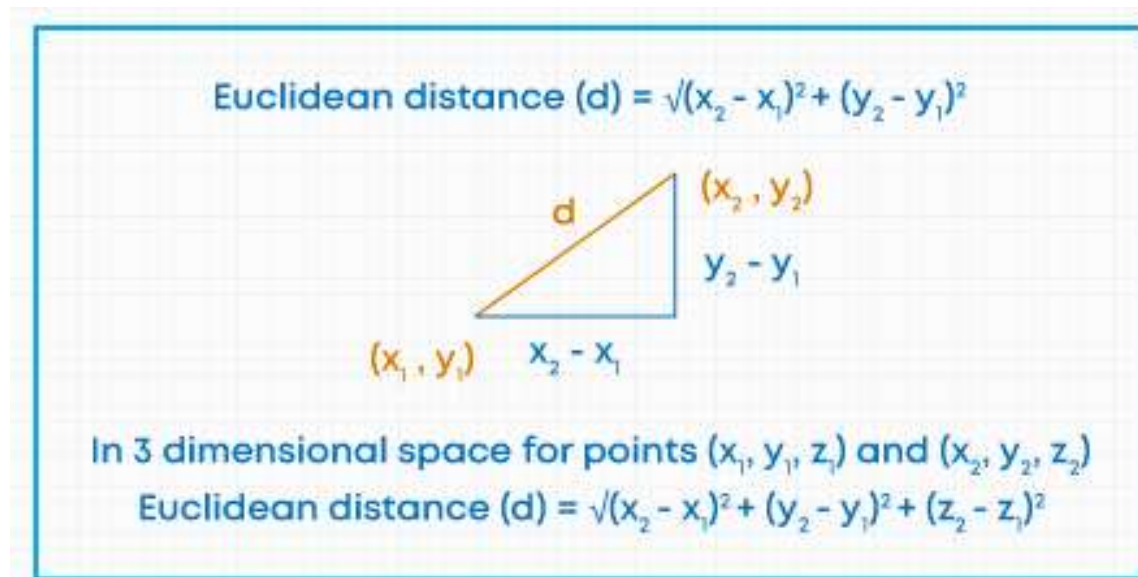


Agglomerative Hierarchical Clustering

- Clustering algorithms use various distance or dissimilarity measures to develop different clusters. Lower/closer distance indicates that data or observation are similar and would get grouped in a single cluster. Remember that the higher the similarity depicts observation is similar.
- Step 2 can be done in various ways to identify similar and dissimilar measures. Namely,
 - ✓ Euclidean Distance
 - ✓ Manhattan Distance
 - ✓ Minkowski Distance
 - ✓ Jaccard Similarity Coefficient
 - ✓ Cosine Similarity
 - ✓ Gower's Similarity Coefficient

Euclidean Distance

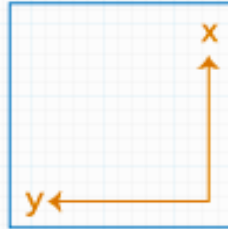
- The Euclidean distance is the most widely used distance measure when the variables are continuous (either interval or ratio scale).
- The Euclidean distance between two points calculates the length of a segment connecting the two points. It is the most evident way of representing the distance between two points.
- The Pythagorean Theorem can be used to calculate the distance between two points, as shown in the figure below. If the points (x_1, y_1) and (x_2, y_2) in 2-dimensional space.



Manhattan Distance

- Euclidean distance may not be suitable while measuring the distance between different locations. If we wanted to measure a distance between two retail stores in a city, then Manhattan distance will be more suitable to use, instead of Euclidean distance.
- The distance between two points in a grid-based on a strictly horizontal and vertical path. The Manhattan distance is the simple sum of the horizontal and vertical components.
- In nutshell, we can say Manhattan distance is the distance if you had to travel along coordinates only.

Manhattan



Euclidean

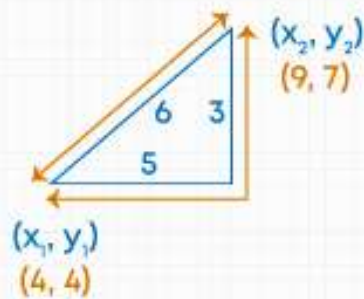


$$\text{Manhattan distance} = \sum_{i=1}^n |x_i - y_i|$$

For 2 dimensional space,

$$M_{\text{dist}} = |x_2 - x_1| + |y_2 - y_1|$$

Example:



Euclidean distance

$$\begin{aligned} &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(9 - 4)^2 + (7 - 4)^2} \\ &= \sqrt{5^2 + 3^2} \\ &= \sqrt{25 + 9} \\ &= \sqrt{34} \\ &= 5.83 \end{aligned}$$

Manhattan distance

$$\begin{aligned} &= |x_2 - x_1| + |y_2 - y_1| \\ &= |9 - 4| + |7 - 4| \\ &= 5 + 3 \\ &= 8 \end{aligned}$$

Minkowski Distance

- Minkowski Distance is the generalized form of Euclidean and Manhattan Distance.
- The formula for Minkowski Distance is given as:

$$D = \left(\sum_{i=1}^n |p_i - q_i|^p \right)^{1/p}$$

- The p parameter of the Minkowski Distance metric of SciPy represents the order of the norm. When the order(p) is 1, it will represent Manhattan Distance and when the order in the above formula is 2, it will represent Euclidean Distance.

Steps to perform Hierarchical Clustering

- We merge the most similar points or clusters in hierarchical clustering – we know this. Now the question is – how do we decide which points are similar and which are not? It's one of the most important questions in clustering!
- Here's one way to calculate similarity – Take the distance between the centroids of these clusters. The points having the least distance are referred to as similar points and we can merge them. We can refer to this as a distance-based algorithm as well (since we are calculating the distances between the clusters).
- In hierarchical clustering, we have a concept called a proximity matrix. This stores the distances between each point. Let's take an example to understand this matrix as well as the steps to perform hierarchical clustering.

Hierarchical Clustering Example

- Suppose a teacher wants to divide her students into different groups. She has the marks scored by each student in an assignment and based on these marks, she wants to segment them into groups.
- There's no fixed target here as to how many groups to have. Since the teacher does not know what type of students should be assigned to which group, it cannot be solved as a supervised learning problem.
- So, we will try to apply hierarchical clustering here and segment the students into different groups.
- Let's take a sample of 5 students:

Student_ID	Marks
1	10
2	7
3	28
4	20
5	35

Creating Proximity Matrix

- First, we will create a proximity matrix which will tell us the distance between each of these points. Since we are calculating the distance of each point from each of the other points, we will get a square matrix of shape $n \times n$ (where n is the number of observations).
- Let's make the 5×5 proximity matrix for our example:

Student_ID	Marks
1	10
2	7
3	28
4	20
5	35



ID	1	2	3	4	5
1	0	3	18	10	25
2	3	0	21	13	28
3	18	21	0	8	7
4	10	13	8	0	15
5	25	28	7	15	0

Creating Proximity Matrix

- The diagonal elements of this matrix will always be 0 as the distance of a point with itself is always 0.
- We will use the Euclidean distance formula to calculate the rest of the distances. So, let's say we want to calculate the distance between point 1 and 2:

$$\sqrt{(10-7)^2} = \sqrt{9} = 3$$

- Similarly, we can calculate all the distances and fill the proximity matrix.

Steps to Perform Hierarchical Clustering

- **Step 1:** First, we assign all the points to an individual cluster:



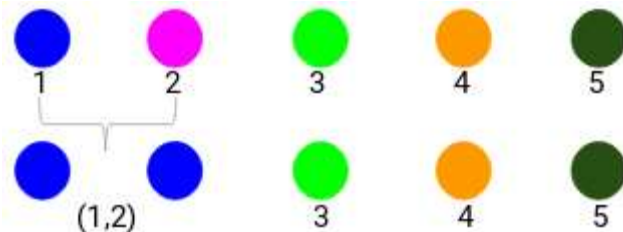
- Different colors here represent different clusters. You can see that we have 5 different clusters for the 5 points in our data.

Steps to Perform Hierarchical Clustering

- **Step 2:** Next, we will look at the smallest distance in the proximity matrix and merge the points with the smallest distance. We then update the proximity matrix:

ID	1	2	3	4	5
1	0	3	18	10	25
2	3	0	21	13	28
3	18	21	0	8	7
4	10	13	8	0	15
5	25	28	7	15	0

- Here, the smallest distance is 3 and hence we will merge point 1 and 2:



Steps to Perform Hierarchical Clustering

- Let's look at the updated clusters and accordingly update the proximity matrix:

Student_ID	Marks
(1,2)	10
3	28
4	20
5	35

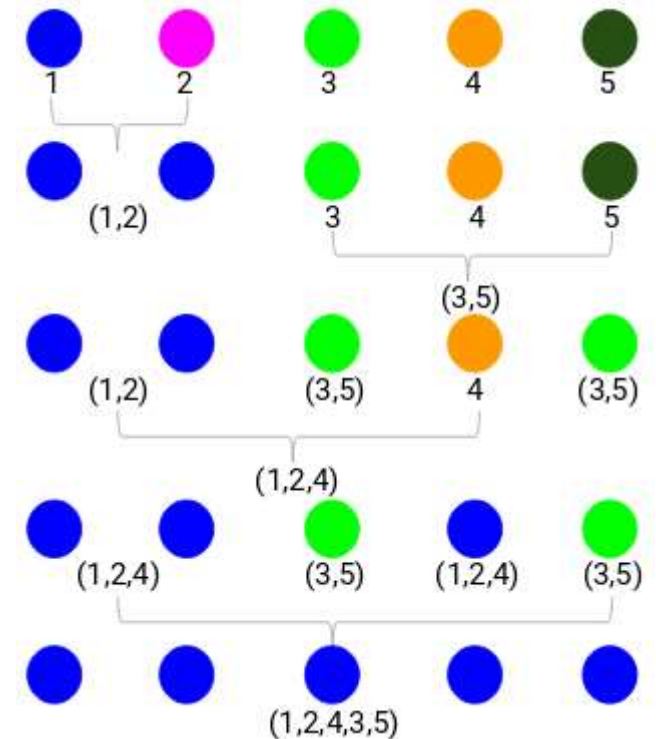
- Here, we have taken the maximum of the two marks (7, 10) to replace the marks for this cluster. Instead of the maximum, we can also take the minimum value or the average values as well. Now, we will again calculate the proximity matrix for these clusters:

ID	(1,2)	3	4	5
(1,2)	0	18	10	25
3	18	0	8	7
4	10	8	0	15
5	25	7	15	0

Steps to Perform Hierarchical Clustering

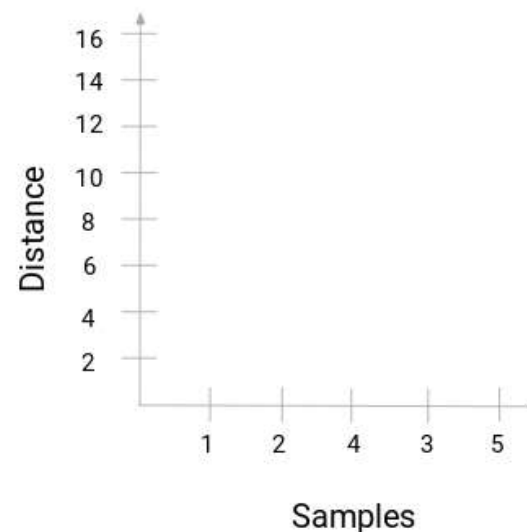
- **Step 3:** We will repeat step 2 until only a single cluster is left.
- So, we will first look at the minimum distance in the proximity matrix and then merge the closest pair of clusters. We will get the merged clusters as shown below after repeating these steps:

- We started with 5 clusters and finally have a single cluster.
- **This is how agglomerative hierarchical clustering works.**



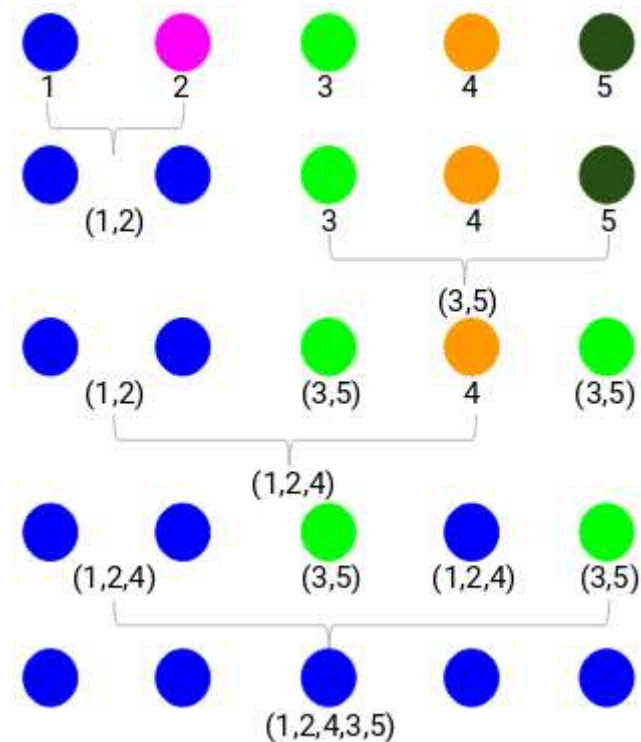
How to choose Number of Clusters in Hierarchical Clustering?

- To get the number of clusters for hierarchical clustering, we make use of an awesome concept called a Dendrogram.
- A dendrogram is a tree-like diagram that records the sequences of merges or splits.
- Let's get back to our teacher-student example. Whenever we merge two clusters, a dendrogram will record the distance between these clusters and represent it in graph form. Let's see how a dendrogram looks like:



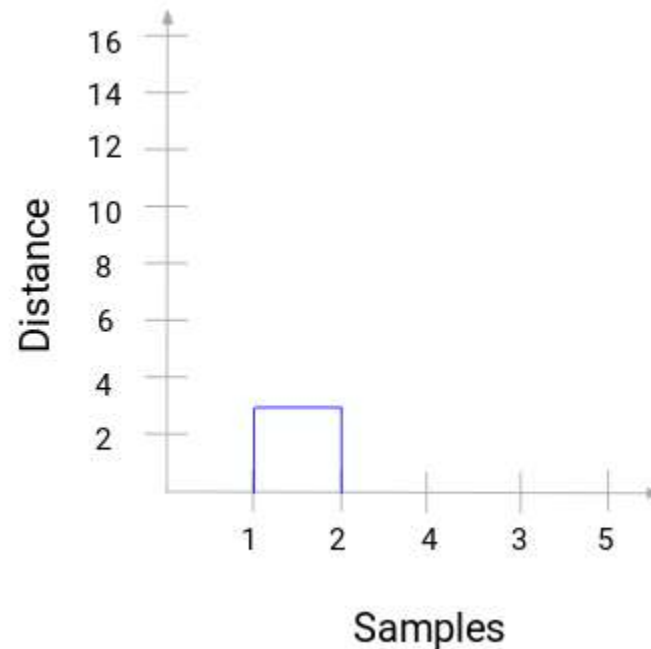
How to choose Number of Clusters in Hierarchical Clustering?

- We have the samples of the dataset on the x-axis and the distance on the y-axis. **Whenever two clusters are merged, we will join them in this dendrogram and the height of the join will be the distance between these points.**
- Let's build the dendrogram for our example:



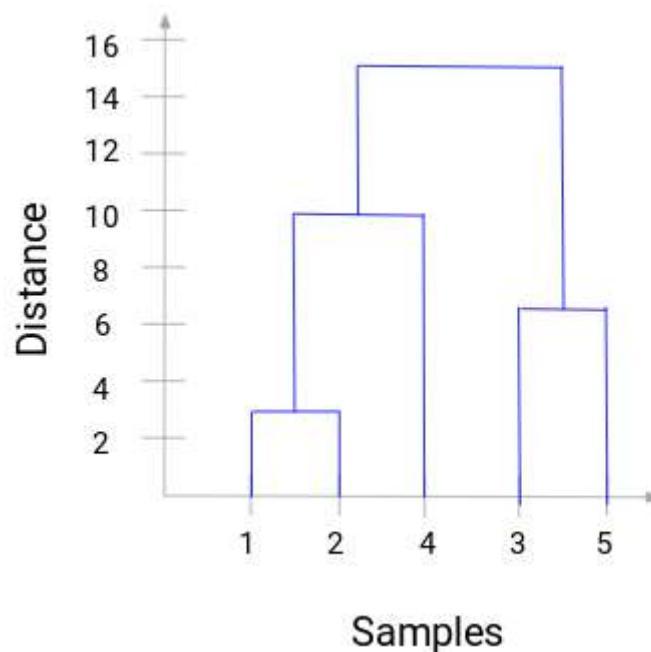
How to choose Number of Clusters in Hierarchical Clustering?

- We started by merging sample 1 and 2 and the distance between these two samples was 3 (refer to the first proximity matrix in the previous section). Let's plot this in the dendrogram:



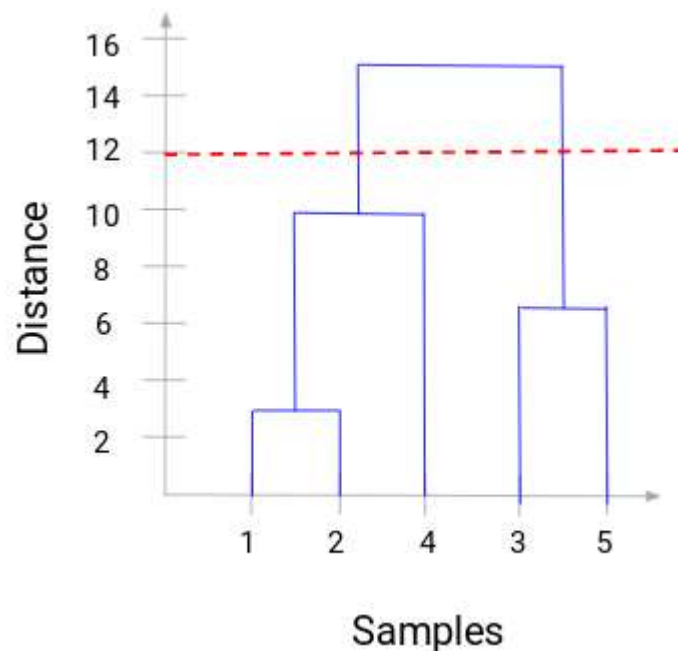
How to choose Number of Clusters in Hierarchical Clustering?

- Here, we can see that we have merged sample 1 and 2. The vertical line represents the distance between these samples. Similarly, we plot all the steps where we merged the clusters and finally, we get a dendrogram like this:



How to choose Number of Clusters in Hierarchical Clustering?

- We can clearly visualize the steps of hierarchical clustering. **More the distance of the vertical lines in the dendrogram, more the distance between those clusters.**
- Now, we can set a threshold distance and draw a horizontal line (Generally, we try to set the threshold in such a way that it cuts the tallest vertical line). Let's set this threshold as 12 and draw a horizontal line:



How to choose Number of Clusters in Hierarchical Clustering?

- The number of clusters will be the number of vertical lines which are being intersected by the line drawn using the threshold.
- In the example, since the red line intersects 2 vertical lines, we will have 2 clusters. One cluster will have a sample (1,2,4) and the other will have a sample (3,5). Pretty straightforward, right?
- This is how we can decide the number of clusters using a dendrogram in Hierarchical Clustering.